temperatures could be lowered sufficiently, the CO_2 in the atmosphere would rain out. There would be a transitional time of CO_2 oceans on Venus. If those oceans could be covered over to prevent re-evaporation—for example, with water oceans made by melting a large, icy moon transported from the outer Solar System—then the CO_2 might conceivably be sequestered away, and Venus converted into a water (or low-fizz seltzer) planet. Ways have also been suggested to convert the CO_2 into carbonate rock.

Thus all proposals for terraforming Venus are still brute-force, inelegant, and absurdly expensive. The desired planetary metamorphosis may be beyond our reach for a very long time, even if we thought it was desirable and responsible. The Asian colonization of Venus that Jack Williamson imagined may have to be redirected somewhere else.

MARS: For Mars we have just the opposite problem. There's not *enough* greenhouse effect. The planet is a frozen desert. But the fact that Mars seems to have had abundant rivers, lakes, and perhaps even oceans 4 billion years ago—at a time when the Sun was less bright than it is today—makes you wonder if there's some natural instability in the Martian climate, something on hair trigger that once released would all by itself return the planet to its ancient clement state. (Let's note from the start that doing so would destroy Martian landforms that hold key data on the past—especially the laminated polar terrain.)

As we know very well from Earth and Venus, carbon dioxide is a greenhouse gas. There are carbonate minerals found on Mars, and dry ice in one of the polar caps. They could be converted into CO_2 gas. But to make enough of a greenhouse effect to generate comfortable temperatures on Mars would require the entire surface of the planet to be plowed up and processed to a depth of kilometers. Apart from the daunting obstacles in practical engineering that this represents—fusion power or no fusion power—and the inconvenience to whatever self-contained, closed ecological